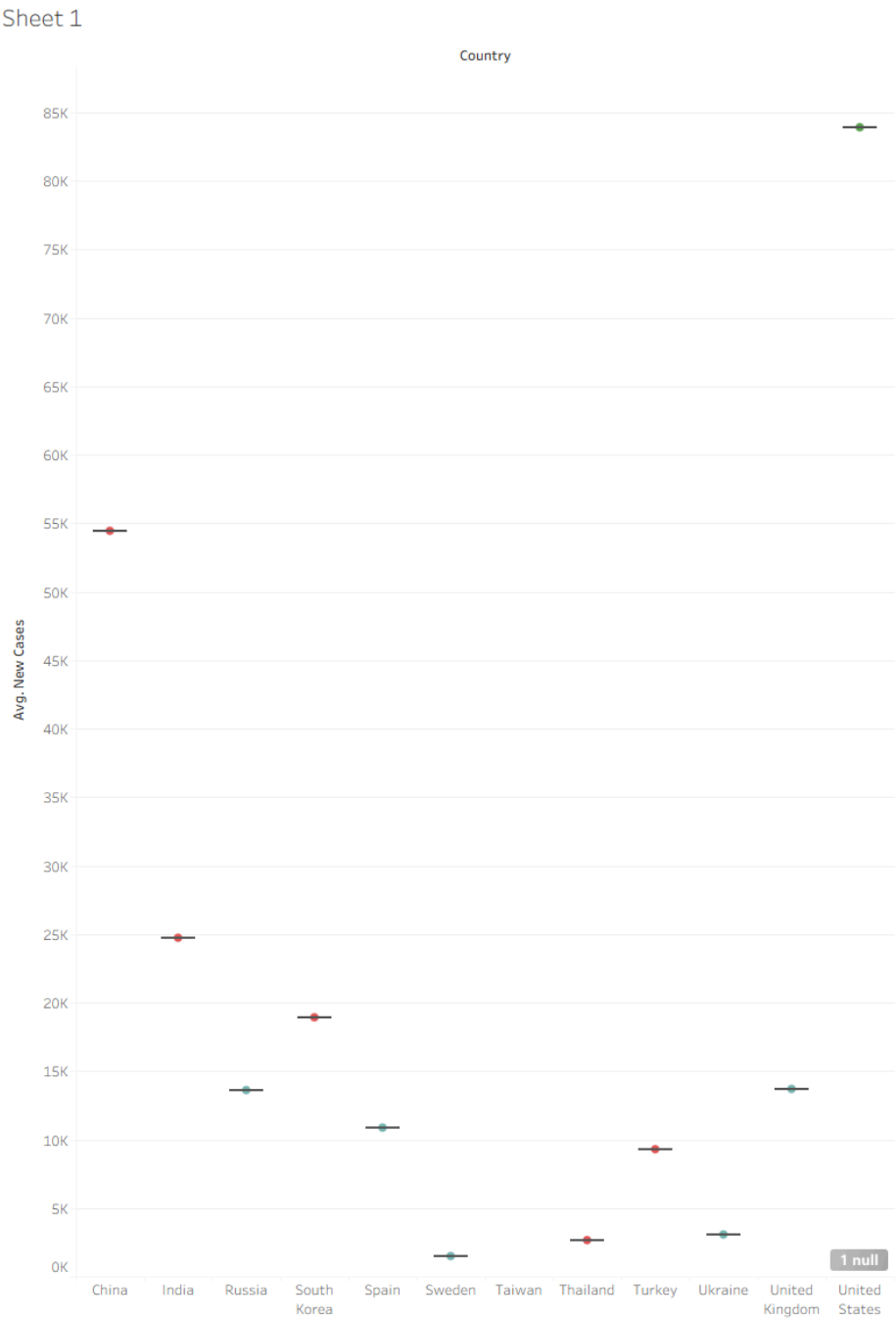


Chart 1: Boxplot (Distribution)



Summary:

This boxplot shows the spread of daily COVID-19 cases across different countries. It highlights which countries had the widest range and the most extreme case counts.

Key Takeaway:

Some countries experienced heavy outliers and greater variability, while others had more stable daily counts.

Accessibility Improvements:

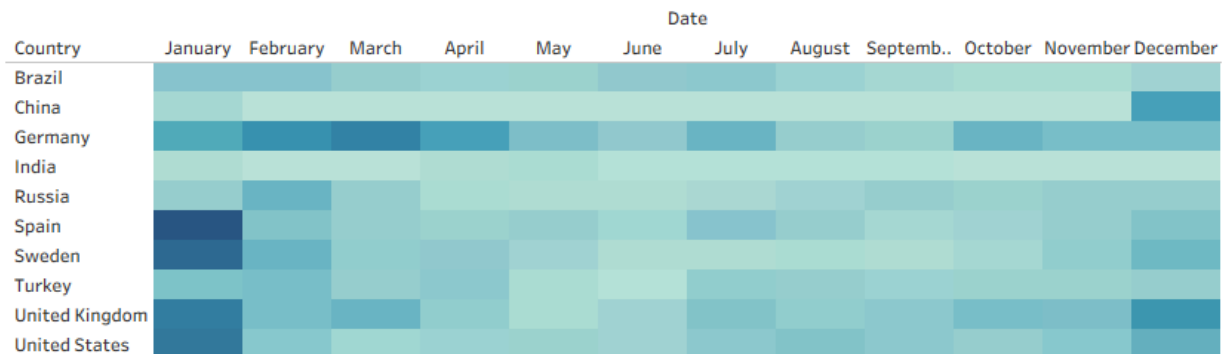
Clear axis labels and a simple, high-contrast color palette were used. Tooltips or hover labels were added (if interactive) to display quartile and outlier values, improving readability for all viewers.

Comparison with Histogram:

A histogram shows the frequency of values but can hide outliers and quartile boundaries. The boxplot makes the data spread and extreme values clearer, using less space and emphasizing variability more effectively.

Chart 2: Heatmap (Relationship)

Sheet 2



Summary:

This heatmap displays the relationship between new daily cases and vaccination rates across countries over time.

Key Takeaway:

Patterns become visible where higher vaccination rates align with fewer new cases, especially in certain time periods.

Accessibility Improvements:

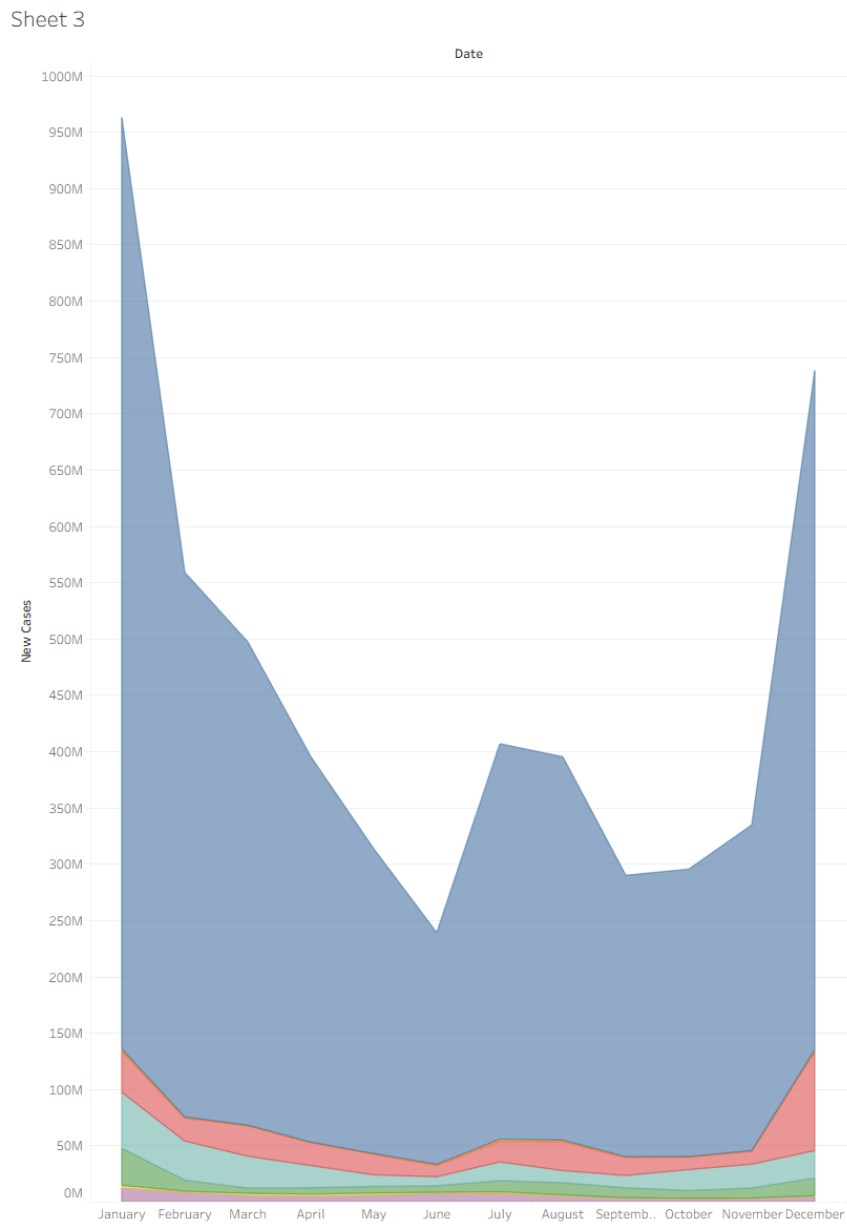
A colorblind-friendly palette (blue to orange) was used. A labeled legend and readable grid lines make it easier to interpret color intensity differences. Text descriptions or captions summarize the key findings for users with limited color perception.

Comparison with Scatterplot:

A scatterplot shows relationships between two variables but struggles with an added time

dimension. The heatmap provides a clear, compact view of how relationships shift over time and across many countries simultaneously.

Chart 3: Area Chart (Composition Over Time)



Summary:

This area chart shows how different COVID-19 variants contributed to total case numbers over time.

Key Takeaway:

It illustrates when certain variants became dominant and how overall case counts shifted with each wave.

Accessibility Improvements:

Each variant has a distinct, labeled color with strong contrast. Peak points are clear, and tooltips (if used interactively) give exact data values.

Comparison with Treemap:

A treemap shows static part-to-whole composition, while the area chart adds a time dimension. This makes it more effective for showing how proportions change dynamically, rather than at a single point in time.